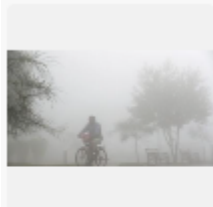


Chemistry

Grade 10

Mixtures and their separations -Past Paper questions

Section A: Select the best choice for each question below.



Q. FOG

answer choices

☐

solution

☐

suspension

☐

colloid

☐

alloy



answer choices

☐

solution

☐

suspension

☐

colloid

☐

alloy



Q. sand in water

answer choices

☐

solution

☐

suspension

☐

colloid

☐

alloy



Q. vinegar

answer choices

- | | |
|--------------------------------|----------------------------------|
| ☐ solution | ☐ suspension |
| ☐ colloid | ☐ alloy |



Q. mayo

answer choices

- | | |
|--------------------------------|----------------------------------|
| ☐ solution | ☐ suspension |
| ☐ colloid | ☐ alloy |



Q. salad dressing

answer choices

- | | |
|--------------------------------|----------------------------------|
| ☐ solution | ☐ suspension |
| ☐ colloid | ☐ alloy |

Q. The part of a solution that does the dissolving

answer choices

- | | |
|---|----------------------------------|
| ☐ solute | ☐ solvent |
| ☐ mixer upper thing | ☐ surfactant |

Q. The part of the solution that is being dissolved

answer choices

- | | |
|----------------------------------|--------------------------------|
| ☐ solute | ☐ solvent |
| ☐ solubility | ☐ miscible |

Q. What is it called when light passes through a colloid and scatters/spreads out

answer choices

☐ tyndall effect

☐ fog effect

☐ spreading out effect

☐ substance effect

Q. If a spoonful of sugar is mixed in a glass of water, what is the water called?

answer choices

☐ solvent

☐ solute

☐ liquid

☐ element

Section B: Answer ALL questions in this section

Question # 1

The diagram (figure 2) shows a setup of a separation technique used in the laboratory.

Answer the following questions, using the diagram below.

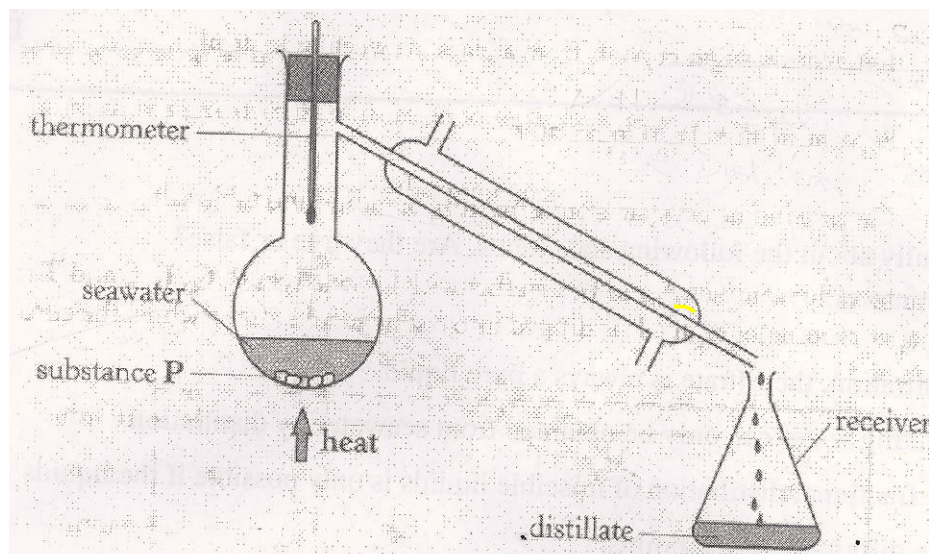


Figure 2

a. Identify substance P. (1 mark)

b. What is the objective of substance P? (1 mark)

c. Identify the distillate collected in the receiver. (1 mark)

- d. Identify the residue. (1 mark)
- e. Suggest how you would determine that the distillate collected is a pure substance. (1 mark)
- f. Label the **water in** and **water out** on the diagram. (2 marks)
- g. Name the method of separation that uses this set of apparatus. (1 mark)
- h. State one possible mistake in the setup of the apparatus shown (figure 2) (1 mark)
- i. State three methods that can be used to determine whether a substance is pure.
- j. What two (2) changes of states takes place during distillation?
- k. What difference in properties is needed to separate two substances by distillation?

Number these steps to give the correct order of the distillation process.
The first one has been done for you.

.....	Pure water evaporates from the mixture to become water vapour.
.....	Water vapour condenses and the water drips into the distillate flask.
1	A mixture of salty water is heated in a boiling flask.
.....	Pure water is collected in the distillate flask.
.....	The water vapour enters the condenser and cools down.

Question # 2

Tara has a sample of sodium chloride which became contaminated by a small amount of sand and iodine crystals.

Plan and design an experiment to remove the iodine and sand from the mixture in order to obtain pure sodium chloride.

(i) Name the method(s) that will be used to separate each solid. (3 mks)

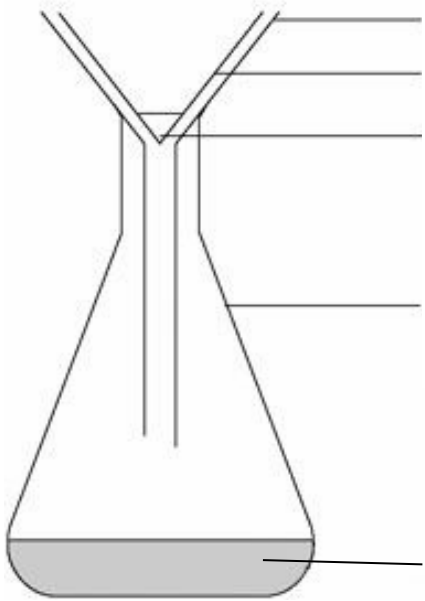
(ii) A suggested list of the apparatus and chemicals which you will use in obtaining the pure solid sample from the mixture. (4 marks)

(iii) An outline of the procedure to be used. (4 marks)

(iv) List the MAIN observations that will be expected at EACH stage of the experiment. (4 mks).

Question # 3

- a. Label the diagram below:



Ryan collects a beaker full of seawater, sand and pebbles from the beach. He wants to separate the pebbles and sand into different containers, and to remove the water to leave behind any dissolved substances.

PRACTICE

- a) How could he separate the pebbles from the rest of the mixture?

.....

.....

- b) Ryan pours the remaining mixture of sand and seawater through filter paper. Explain how this step works, and say what would be in the residue and the filtrate.

.....

.....

Residue: Filtrate:

- c) He then uses an evaporating basin and Bunsen burner to heat the filtrate. Explain what happens in this step, and what will be left at the end.

.....

.....

.....

Question # 4

Solid, gaseous and liquid substances combine in various ways to form mixtures. The properties of mixtures depend on those of the substances that make them up. These properties also serve as a means by which mixtures may be classified base on how they may be separated.

a. (i) Suspensions, solutions and colloids are examples of mixtures. Distinguish among solutions, suspensions and colloids in terms of, sedimentation, passage of light and size of dispersed particles. (9 marks)

(ii) Draw a labelled diagram showing the comparison of the particle sizes in a solution, colloid and suspension. (4 mks)

b. What are the main differences between pure substances and mixtures (3 marks)

c. Distinguish between homogeneous and heterogeneous mixtures. Give two examples of each.
(6 marks)